Course: Introduction to Artificial Intelligence
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MapOut – AI-Powered Travel Planner

Week 6: Constraint Satisfaction Problems and Planning

This week's planned objective

Implement a CSP-based itinerary planner using backtracking, support multi-day trips, and integrate real travel distances and times while respecting user preferences and constraints.

What I completed

- Implemented a CSP planner using backtracking.
- Added destination, interests, budget, daily time, and travel time constraints.
- Implemented multi-day itinerary generation without duplicate attractions.
- Integrated OpenRouteService for real walking and driving routes.
- Added route caching to reduce repeated API requests.
- Organized the project and successfully tested the implemented functionality.

Evidence submitted

- **csp_planner.py** – CSP implementation and backtracking.
- **route_matrix.py** – real travel time and distance calculations.
- **main.py** – generated multi-day itinerary.
- Test results and screenshots demonstrating successful execution.
- Week 6 report explaining the implementation and results.

What I can demonstrate now

I can demonstrate how MapOut generates a three-day itinerary for Ohrid based on a selected budget, interests, available time, and transportation mode. I can also demonstrate real walking and driving route calculations and show that the CSP planner satisfies the tested constraints.

Problem or failure encountered

During integration, the validator and CSP planner used different formats for attraction categories, which caused an error. I corrected the category handling and successfully reran the application.

Decision made this week

I decided to use CSP with backtracking for multi-day itinerary generation and OpenRouteService for real travel calculations. I also separated dataset validation from the main application to keep the output focused on the generated itinerary.

AI assistance used

I used ChatGPT to understand CSP and backtracking, receive Python implementation guidance, debug errors, organize tests, and improve documentation. I reviewed and tested the suggested solutions.

Feedback received

To be completed after receiving feedback.

Next week's objective

Improve the itinerary planner by considering accommodation, restaurants, and nearby locations to create more complete and realistic travel plans..